

OptiCrop: Performance Testing

Model Performance (Test Set)

Metric	Score
Accuracy	97.27%
Precision	97.42%
Recall	97.27%
F1-Score	97.25%
ROC-AUC	0.9998

Functional Testing (Web App)

Test Case	Steps	Expected Result	Status
Valid input submission	Enter valid N,P,K,temp,humidity, Return a single predicted crop name	Return a single predicted crop name	Pass
Home page loads	Navigate to /	Input form (index.html) renders correctly	Pass
Model artifact loading	Start app.py	model.pkl, scaler.pkl, label_encoder.pkl load without error	Pass

Response Time Observation

Since the model is a lightweight Logistic Regression classifier (no deep learning inference), prediction requests return in well under 1 second on standard hardware, with no noticeable lag for a single request.

Known Limitations

- No automated load-testing (e.g. Locust) has been run in this version; response-time notes above are based on manual local testing only.
- Testing was performed locally; behavior under concurrent multi-user load has not been separately measured.